M6-L1 UNIVERSAL CONTROLLER / ENGINEERING CANDIDATE
ESP32-S3-WROOM-1-N4 / RV-3032-C7 / isolated RF80 + RF2M contracts
No manufacturing or operational release. All physical tests NOT_RUN.
LoRa optional433/868 nominal package/RF-feed audit M6-L1: trusted assembly configuration required. Physical tests NOT_RUN.

01 power / 1

File: 01_power.kicad_sch

01 power / 2

File: 01_power_2.kicad_sch

02 USB reset / 1

File: 02_USB_reset.kicad_sch

02 USB reset / 2

File: 02_USB_reset_2.kicad_sch

03 MCU / 1

File: 03_MCU.kicad_sch

04 RTC analog / 1

File: 04_RTC_analog.kicad_sch

05 static safety / 1

File: 05_static_safety.kicad_sch

05 static safety / 2

File: 05_static_safety_2.kicad_sch

05 static safety / 3

File: 05_static_safety_3.kicad_sch

06 RF80 / 1

File: 06_RF80.kicad_sch

06 RF80 / 2

File: 06_RF80_2.kicad_sch

07 RF2M / 1

File: 07_RF2M.kicad_sch

07 RF2M / 2

File: 07_RF2M_2.kicad_sch

08 status / 1

File: 08_status.kicad_sch

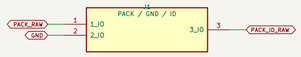
09 LoRa / 1

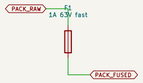
File: 09_LoRa.kicad_sch

10 test / 1

File: 10_test.kicad_sch

ARDF-TX / NOT RELEASED
Sheet: /
File: CTRL.kicad_sch
Title: ARDF-TX universal ESP32-S3 controller
Size: A2 | Date: 2026-10-18 | Rev: M6-C1R1 candidate
KiCad E.D.A. 10.0.6 | Id: 1/17











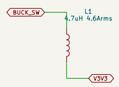










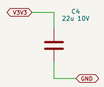


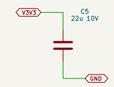


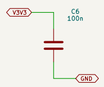


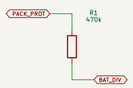


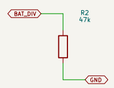


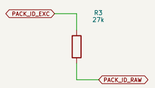



M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

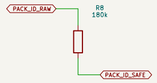
**ARDF-TX**

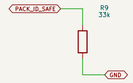
Sheet: /01 power / 1/
File: 01_power.kicad_sch

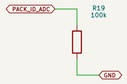
***Title: 01 power / 1***

| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
| KiCad E.D.A. 10.0.6 | | Id: 2/17 |



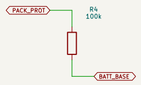



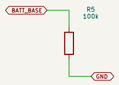



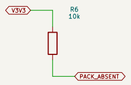


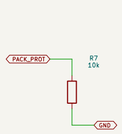





M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

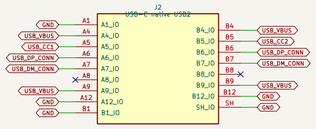
Evaluation review only: no ordering or manufacture

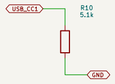
**ARDF-TX**

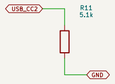
Sheet: /01 power / 2/
File: 01_power_2.kicad_sch

***Title: 01 power / 2***

| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/17 |

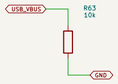


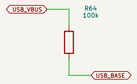


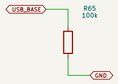


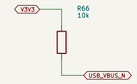


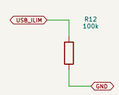


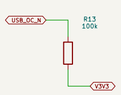


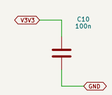


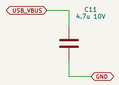



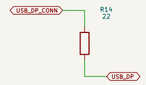



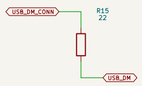





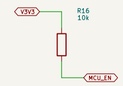











M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

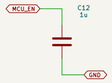
Evaluation review only: no ordering or manufacture

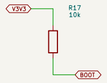
**ARDF-TX**

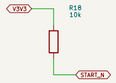
Sheet: /02 USB reset / 1/
File: 02_USB_reset.kicad_sch

***Title: 02 USB reset / 1***

| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/17 |
















M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

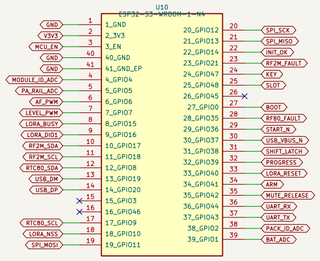
Evaluation review only: no ordering or manufacture

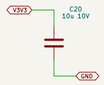
**ARDF-TX**

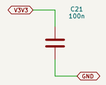
Sheet: /02 USB reset / 2/
File: 02_USB_reset_2.kicad_sch

***Title: 02 USB reset / 2***

| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/17 |







M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

**ARDF-TX**

Sheet: /03 MCU / 1/
File: 03_MCU.kicad_sch

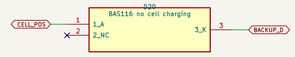
***Title: 03 MCU / 1***

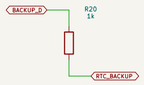
| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
| KiCad E.D.A. 10.0.6 | | Id: 6/17 |

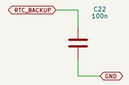


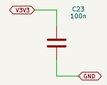


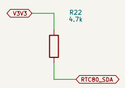


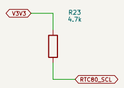


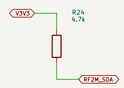



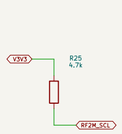



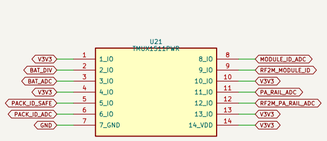



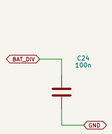



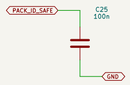



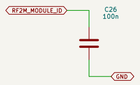



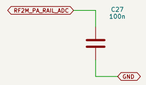



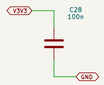



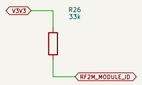




M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

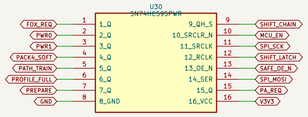
Evaluation review only: no ordering or manufacture

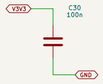
**ARDF-TX**

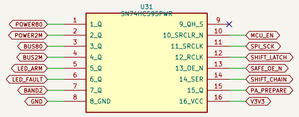
Sheet: /04 RTC analog / 1/
File: 04_RTC_analog.kicad_sch

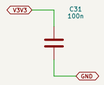
***Title: 04 RTC analog / 1***

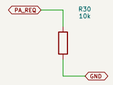
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/17 |

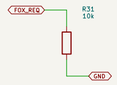


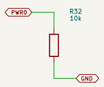



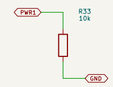



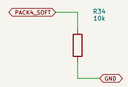



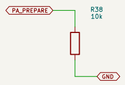


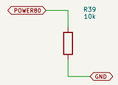


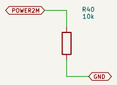


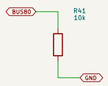


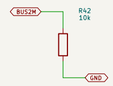


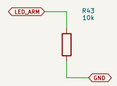


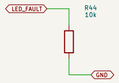



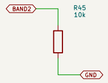



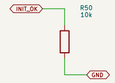

M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

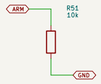
Evaluation review only: no ordering or manufacture

| | | |
|---|---|---|
| **ARDF-TX** | | |
| Sheet: /05 static safety / 1/ | | |
| File: 05_static_safety.kicad_sch | | |
| ***Title: 05 static safety / 1*** | | |
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
| KiCad E.D.A. 10.0.6 | | Id: 8/17 |



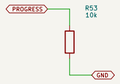



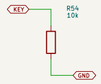



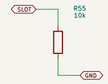



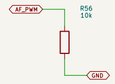














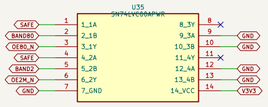






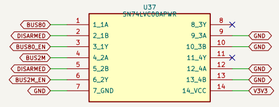











M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

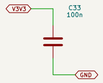
**ARDF-TX**

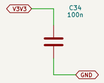
Sheet: /05 static safety / 2/
File: 05_static_safety_2.kicad_sch

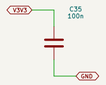
***Title: 05 static safety / 2***

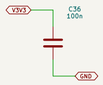
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 9/17 |

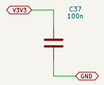


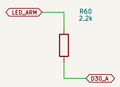



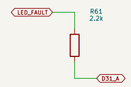



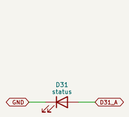



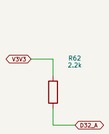



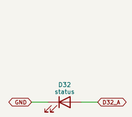







M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

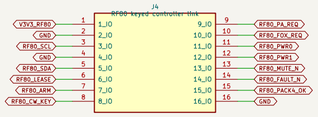
Evaluation review only: no ordering or manufacture

**ARDF-TX**

Sheet: /05 static safety / 3/
File: 05_static_safety_3.kicad_sch

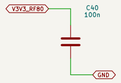
***Title: 05 static safety / 3***

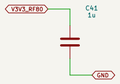
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 10/17 |

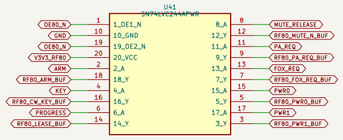


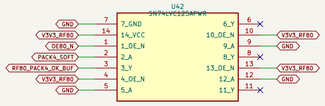


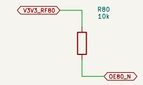


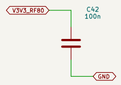


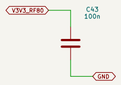



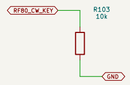


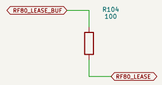


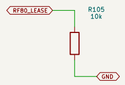


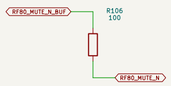


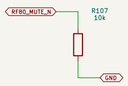


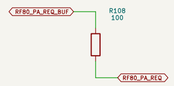



M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

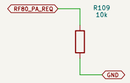
Evaluation review only: no ordering or manufacture

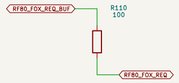
**ARDF-TX**

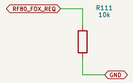
Sheet: /06 RF80 / 1/
File: 06_RF80.kicad_sch

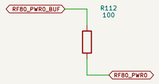
***Title: 06 RF80 / 1***

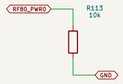
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 11/17 |

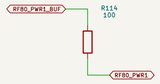


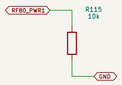



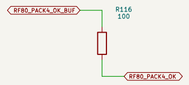



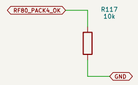



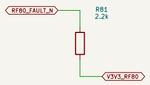













M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

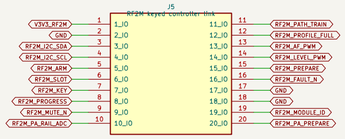
Evaluation review only: no ordering or manufacture

**ARDF-TX**

Sheet: /06 RF80 / 2/
File: 06_RF80_2.kicad_sch

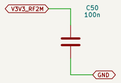
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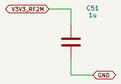
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 12/17 |

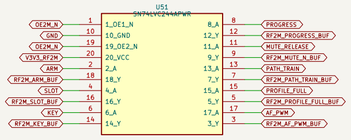


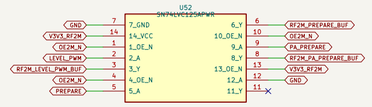


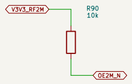


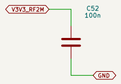


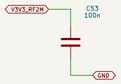



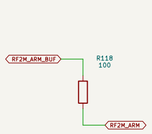



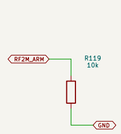



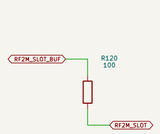



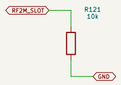



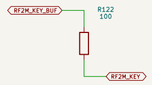



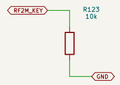



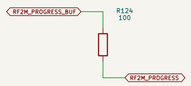



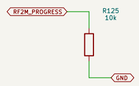



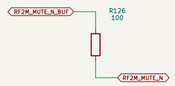



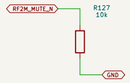



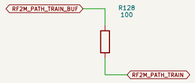



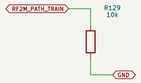

M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

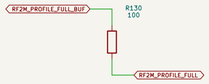
Evaluation review only: no ordering or manufacture

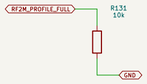
| | | |
|---|---|---|
| **ARDF-TX** | | |
| Sheet: /07 RF2M / 1/ | | |
| File: 07_RF2M.kicad_sch | | |
| ***Title: 07 RF2M / 1*** | | |
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
| KiCad E.D.A. 10.0.6 | | Id: 13/17 |

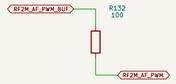


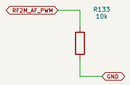



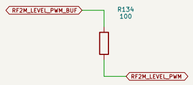



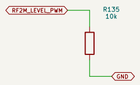



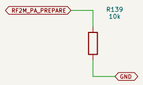


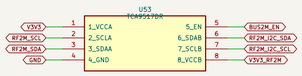


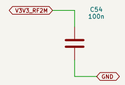






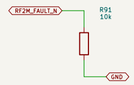









M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

**ARDF-TX**

Sheet: /07 RF2M / 2/
File: 07_RF2M_2.kicad_sch

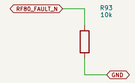
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| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 14/17 |










M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

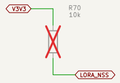
**ARDF-TX**

Sheet: /08 status / 1/
File: 08_status.kicad_sch

***Title: 08 status / 1***

| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
| KiCad E.D.A. 10.0.6 | | Id: 15/17 |
















M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

**ARDF-TX**

Sheet: /09 LoRa / 1/
File: 09_LoRa.kicad_sch

***Title: 09 LoRa / 1***

| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 16/17 |































M6-C1R1 candidate: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

**ARDF-TX**

Sheet: /10 test / 1/
File: 10_test.kicad_sch

***Title: 10 test / 1***

| Size: A2 | Date: 2026-10-18 | **Rev: M6-C1R1 candidate** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 17/17 |